AI for Software Architecture

by team CODECRAFTERS

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Research Synthesis

How AI augments architectural decision-making across the full lifecycle

What Makes Software Architecture Hard?

The Core Challenge

Software architecture requires balancing **complex trade-offs:**

- Scalability vs. Cost
- Performance vs. Maintainability
- Speed vs. Reliability

Key Insight

Architecture decisions are expensive to change later.

Two Types of Requirements

FR (Functional)
What the system does

↓

NFR (Non-Functional)
How well it does it

Architecture is primarily driven by NFRs

Technical Debt: The Cost of Bad Architecture

What is Technical Debt?

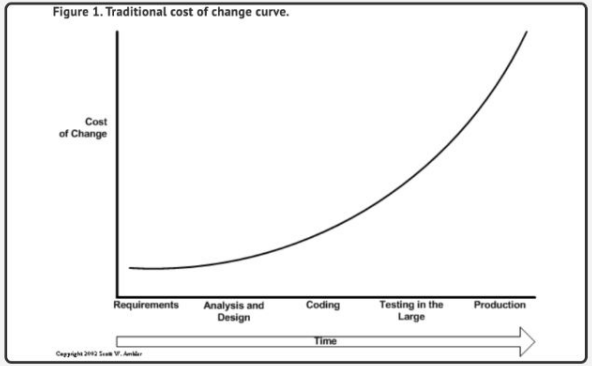
The **accumulated cost** of poor architectural decisions over time.

Examples

- **Tight coupling** → hard to scale
- **Monolith** → slows development
- **No separation of concerns** → fragile changes

"Bad architecture = exponential technical debt"

Technical Debt Growth Over Time



Reference

Where Do Humans Struggle?

Human Limitations

- Limited ability to **explore large design spaces**
- Cannot **predict future system behavior** accurately

- Past experience & intuition
- Rules of thumb (heuristics)
- Trial & error

Bridge to AI

"Architecture is a decision problem under uncertainty."

Requirements (Ambiguous)

↓

Human Architect

↓

Experience + Heuristics

↓

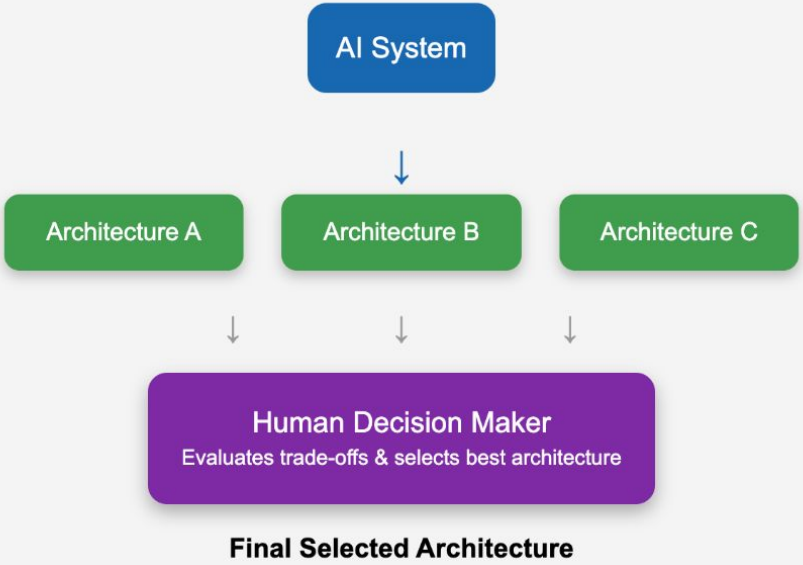
One-Shot Design Guess

↓

High Risk of Error

AI-Assisted Architecture Decision Making

AI generates **multiple candidate architectures**, enabling humans to make informed decisions based on analysis.



Key Findings: GenAI Applications and Challenges

- **Usage Gap:** Predominantly used for implementation (docs/code) rather than high-stakes architectural design.
- **Legacy Modernization:** Actively aids in analyzing monolithic systems and migrating them to microservices.
- **Assistive Role:** Functions as a supportive tool rather than a fully autonomous decision-maker.
- **Black Box:** AI in general lacks context.

Azure Advisor

Advisor | Reliability

Search

Overview
Getting started
Advisor score
Workbooks
Assessments (Preview)

Recommendations

Cost
Security
Reliability
Operational excellence
Performance
All recommendations

Monitoring

Alerts (Preview)
Recommendation digests

Settings

Configuration

Support + troubleshooting

Support + troubleshooting

Feedback | Download as CSV | Download as PDF | Create alert | Create recommendation digest | Workbooks

Subscription equals 27 of 38 selected | Recommendation Status equals Active | Resource Group equals All | Type equals All | Add filter

No grouping

Total recommendations: 5

Recommendations by impact: 2 High impact, 2 Medium impact, 1 Low impact

Impacted resources: 56

To get additional reliability recommendations, review Advisor reliability workbook

Impact	Description	Potential benefits	Impacted resources	Last updated
High	Use Availability zones for better resiliency and availability	Usage of zonal VMs protect your apps from zonal outage in any oth...	5 Virtual machines	3/6/2024, 11:00 AM
High	Create an Azure Service Health alert	Stay informed about issues and advisories across 4 areas (Service...	1 Subscription	3/6/2024, 04:25 PM
Medium	Enable Cross Region Restore for your recovery Services Vault	As one of the restore options, Cross Region Restore (CRR) allows you t...	37 Recovery Services va...	3/6/2024, 06:14 PM
Medium	Use NAT gateway for outbound connectivity	Prevent outbound connection failures with NAT gateway	8 Virtual networks	3/6/2024, 04:38 PM
Low	Use Service Bus premium tier for improved resilience	Service Bus premium tier offers better resiliency with CPU and...	5 Service bus namespaces	3/6/2024, 06:34 PM

Showing 1 - 5 of 5 results.

Are these recommendations helpful?

Suggests Changes that

- Reduces cost
- Increases reliability
- Reduces storage

Image credits: Azure doc

Experiment: Choosing the right architecture using AI

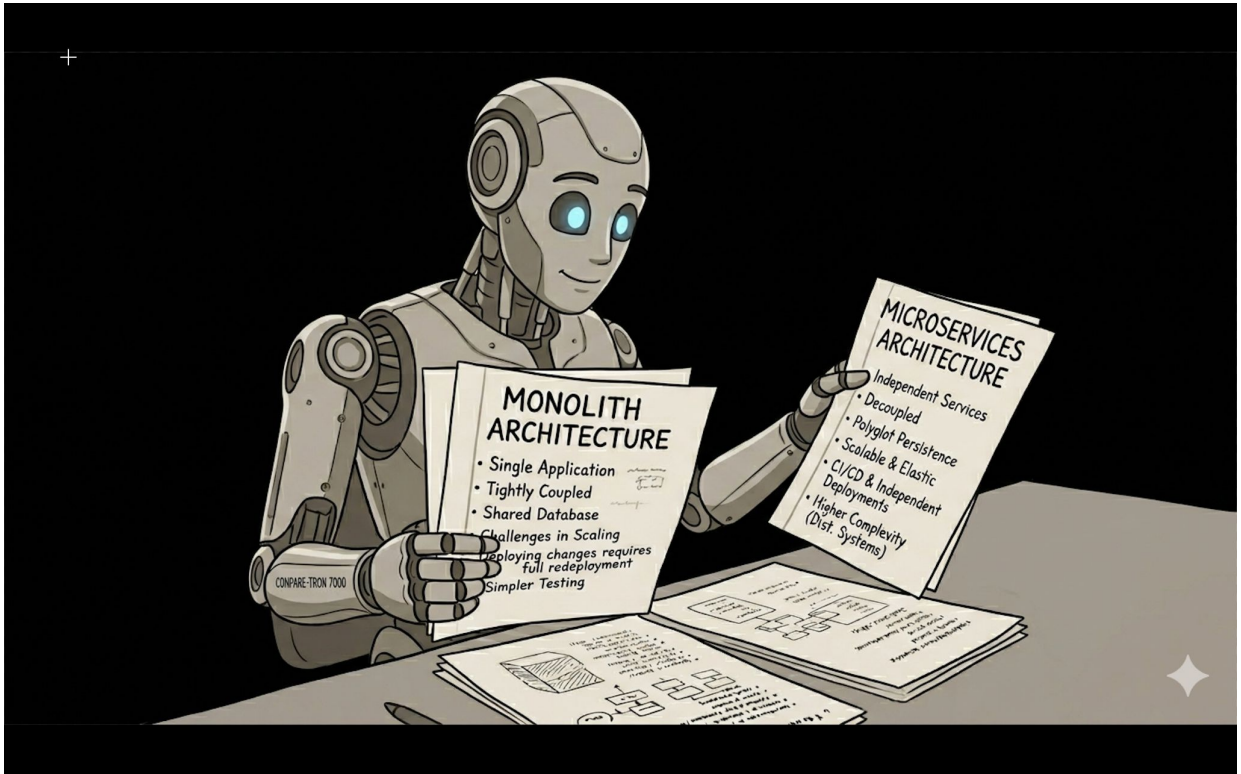
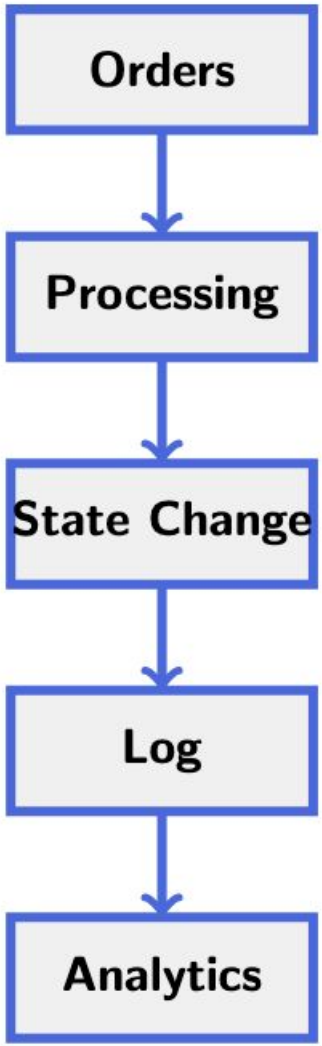


Image credits: Gemini

What are we making?

We want the
perfect architecture



Zero-Shot Prompt:

"Design a backend system for a high-throughput order matching engine with:

- Continuous order stream processing
- Real-time matching (buy/sell orders)
- Support for 10K-1M orders
- Low latency requirements
- Metrics tracking (arrival, processing, latency)
- Small team implementability

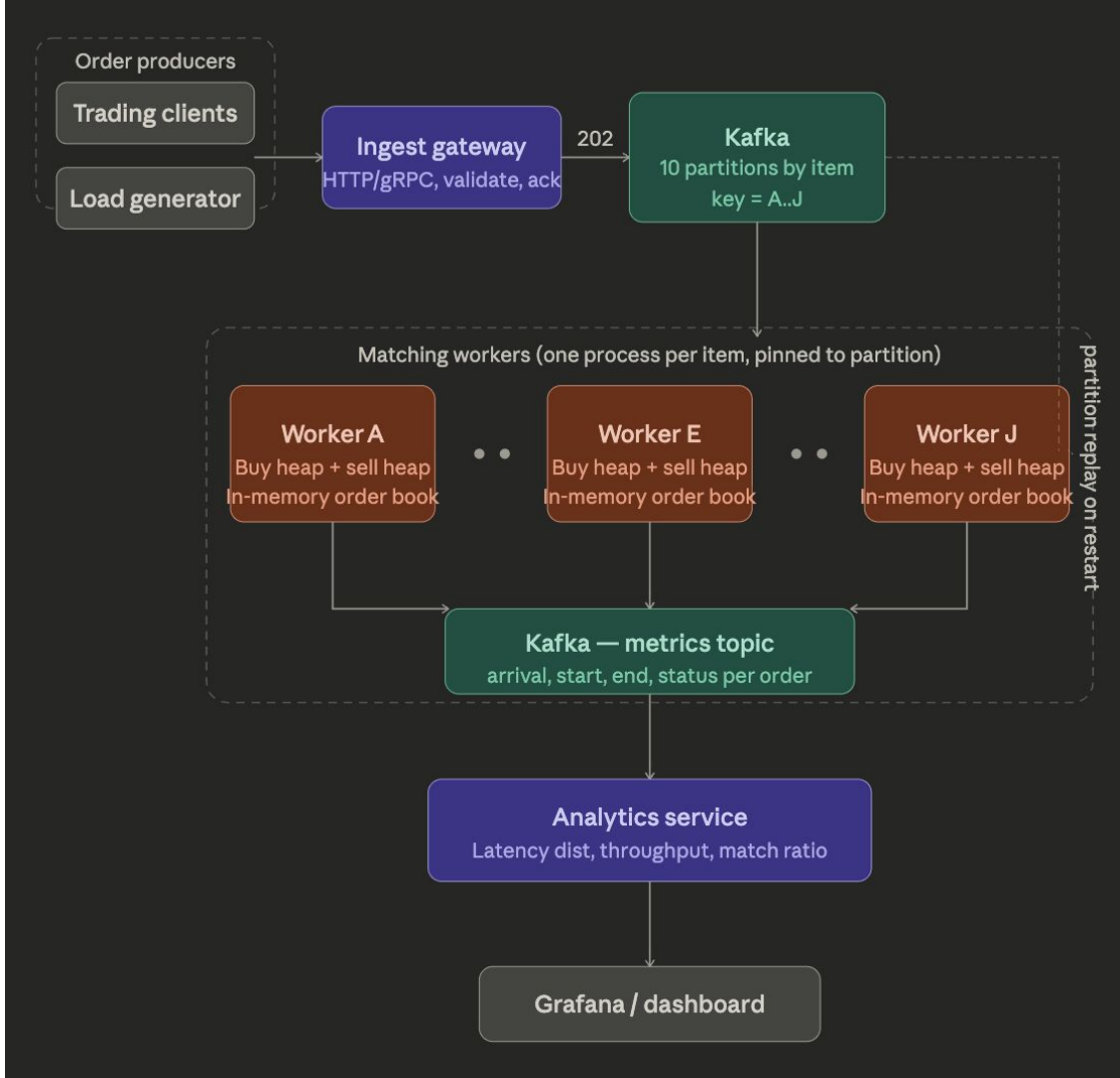
Propose multiple architectures with trade-off analysis."

Output Format

1. Component breakdown
2. Data flow diagrams
3. Trade-off analysis
4. Estimated Performance metrics

Image credits: Gemini

What does Claude say?



AI for Software Architecture

Async Worker-Based System (Claude Recommendation)

Research Synthesis

Client → Ingest Gateway → Kafka (Queue) → Workers → Metrics / Output

1. Decoupled Ingestion: Orders are instantly written to Kafka, separating request handling from processing.

2. Parallel Workers (No Locks): Each worker handles a partition → no shared state → no locking → high throughput.

3. Replayable State: Workers rebuild state from Kafka logs → durability without database bottlenecks.

ASYNC WORKERS
MODULAR MONOLITH
EVENT DRIVEN MICROSERVICES
SILVER
GOLD
BRONZE
Image credits: Gemini

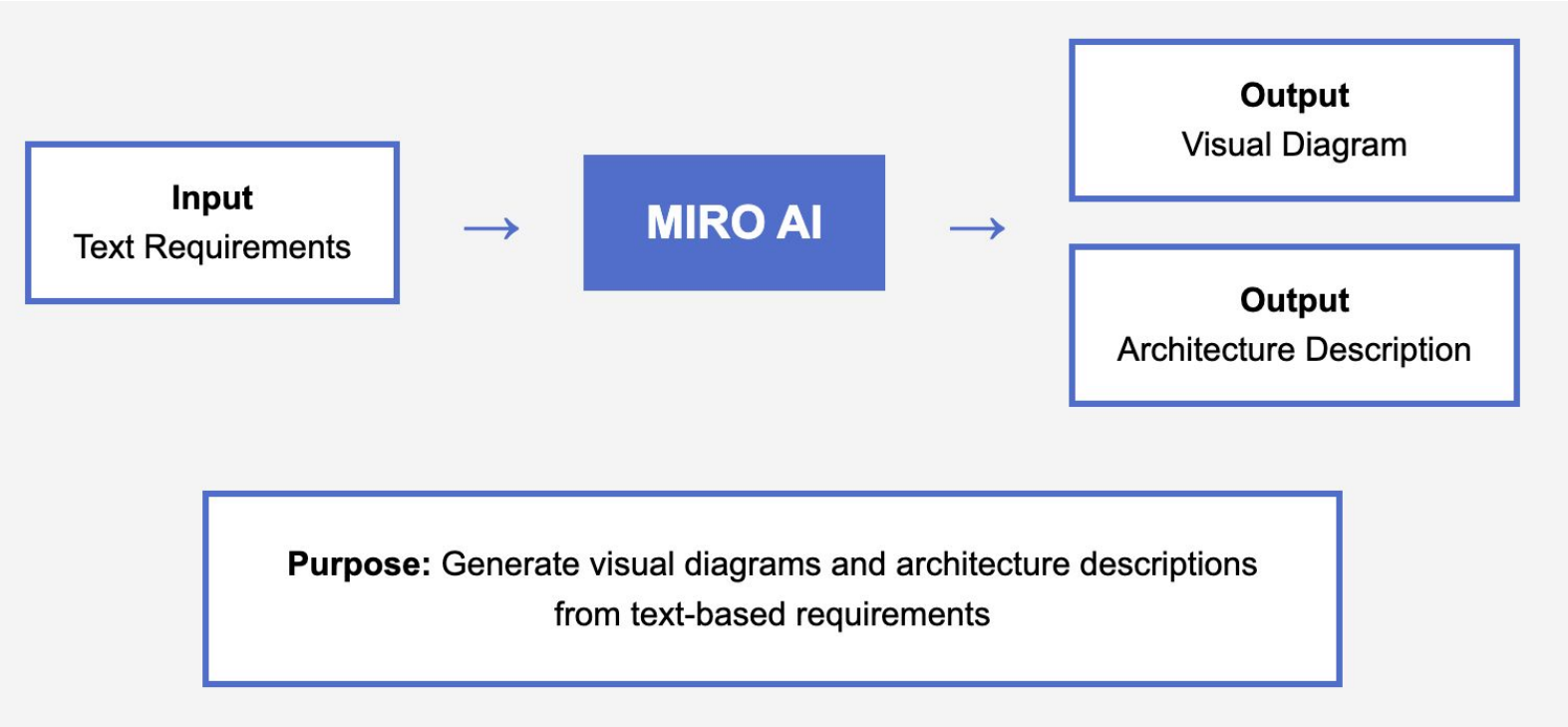
RECOMMENDED ARCHITECTURE

Asynchronous Worker-Based System

Consensus from 4 GenAI models

Dimension	Async Worker-Based	Modular Monolith	Event-Driven Architecture	Microservices Architecture
Latency	Medium (queue adds slight delay)	Lowest (in-process calls)	Medium–High (event propagation delay)	High (multiple network hops)
Throughput	High (parallel workers + queue)	Medium (limited by single machine)	High (stream processing + parallel consumers)	Very High (independent service scaling)
Scalability	Horizontal (add more workers)	Low (mostly vertical scaling)	Very High (partitioned event streams)	Very High (independent scaling per service)
Complexity	Medium (queue + worker coordination)	Low (single codebase)	High (event coordination, debugging)	Very High (distributed system overhead)
Fault Tolerance	High (queue retries, worker isolation)	Low (single point of failure)	Very High (replay, decoupled services)	High (service isolation, coordination needed)

MIRO AI


Input
Text Requirements
MIRO AI
Output
Visual Diagram
Output
Architecture Description
Purpose: Generate visual diagrams and architecture descriptions from text-based requirements

Apr 1 1:19 AM

miro ai for making system d | My First Board - Miro

miro.com/app/board/uXjVGpYOGCA=/

miro | My First Board | Upgrade | Present | Share

Hey Umang, what are we working on today?

I want to create a system design for matching engine for highly scalable and reliable .

Brainstorm ideas | Map user journey | Plan roadmap | Create research plan | Create requirements doc

Explore Flows | Explore Sidekick

100%

Apr 1 1:23 AM
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My First Board - Miro
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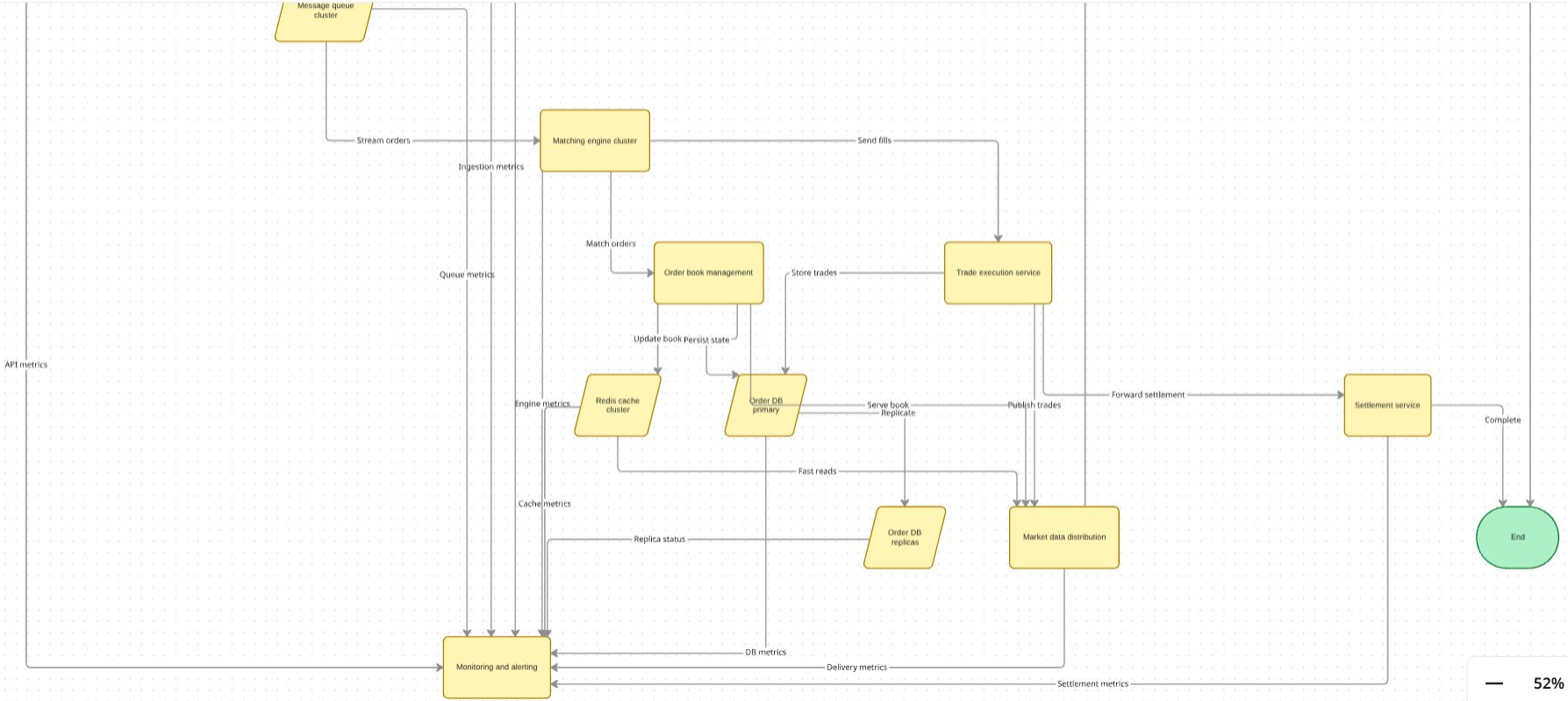
miro My First Board
Back to canvas
Share
Diagram
Layers



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Conclusions

- **AI augments decision-making**, but does not replace human architects — it generates options, humans choose.
- **Async worker-based architecture** emerges as the most practical and scalable solution for our case.
- **Human-in-the-loop is critical** because AI can hallucinate and produce incorrect architectural decisions.

References

[1] Alessio Bucaioni, Martin Weyssow, Junda He, Yunbo Lyu, and David Lo. Artificial intelligence for software architecture: Literature review and the road ahead. *arXiv preprint arXiv:2504.04334*, 2025.

[2] Rudra Dhar, Karthik Vaidhyanathan, and Vasudeva Varma. Can llms generate architectural design decisions?—an exploratory empirical study. In *2024 IEEE 21st International Conference on Software Architecture (ICSA)*, pages 79–89. IEEE, 2024.

[3] Tobias Eisenreich, Sandro Speth, and Stefan Wagner. From requirements to architecture: An ai-based journey to semi-automatically generate software architectures. In *Proceedings of the 1st International Workshop on Designing Software*, pages 52–55, 2024.

[4] Matteo Esposito, Xiaozhou Li, Sergio Moreschini, Noman Ahmad, Tomas Cerny, Karthik Vaidhyanathan, Valentina Lenarduzzi, and Davide Taibi. Generative ai for software architecture: applications, challenges, and future directions. *Journal of Systems and Software*, page 112607, 2025.

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